APTITUDE

Quantitative Aptitude Guide & Formula Sheet

Formula Handbook & High-Yield Shortcuts

1. Arithmetic & Progression (AP & GP)

- AP nth term: Tn = a + (n - 1)d; Sum of n terms: Sn = n/2 [2a + (n - 1)d].
- GP nth term: Tn = a * r^(n - 1); Sum of n terms: Sn = a(r^n - 1)/(r - 1) for r > 1.
- Infinite GP sum: S_inf = a / (1 - r), where |r| < 1.
- Harmonic Mean (HM) = 2ab / (a + b) for two numbers a and b.

2. Algebra & Logarithms

- (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca).
- a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca).
- If a + b + c = 0, then a^3 + b^3 + c^3 = 3abc.
- log_b(x * y) = log_b(x) + log_b(y); log_b(x / y) = log_b(x) - log_b(y); log_b(x^k) = k * log_b(x).

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Geometry, Mensuration & Probability

3. Mensuration 2D & 3D Cheat Sheet

- Circle Area = π * r^2; Circumference = 2π * r; Sector Area = (θ / 360) * π * r^2.
- Cylinder Volume = π * r^2 * h; Curved Surface Area = 2π * r * h; Total Surface = 2πr(r + h).
- Cone Volume = (1/3) * π * r^2 * h; Slant height l = sqrt(r^2 + h^2).
- Sphere Volume = (4/3) * π * r^3; Surface Area = 4π * r^2.

4. Probability & Combinatorics

- Permutation: nPr = n! / (n - r)! (Arrangement matters).
- Combination: nCr = n! / (r! * (n - r)!) (Selection without ordering).
- Probability P(E) = n(E) / n(S); P(A U B) = P(A) + P(B) - P(A ∩ B).